Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent institution of MAHE, Manipal)

III SEMESTER B.TECH. (BIOTECHNOLOGY)

END SEMESTER EXAMINATIONS

SUBJECT: Bioprocess Calculations [BIO 2152]

REVISED CREDIT SYSTEM

Time: 50 Minutes MAX.30

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitable assumed.

1	Which of the following is the correct representation of a general balance equation? Input – Output – Generation – Consumption = Accumulation Input – Output + Generation – Consumption = Accumulation Input – Output + Generation + Consumption = Accumulation Input – Output – Generation + Consumption = Accumulation Ans: Input – Output + Generation – Consumption = Accumulation	**CO3**	**1**
2	If 200g of 45.0% methanol in water is combined with 300 g of 70.0% methanol in water, what is the composition of the product? 40% water 57.5% water 60% water 42.5% water Ans: 40% water	**CO1**	**6**
3	**The combustion equations of carbon and carbon monoxide are as follows:** $C + O_2 \rightarrow CO_2$; $\Delta H = -394$ kJ/kmol $CO + 0.5O_2 \rightarrow CO_2$; $\Delta H = -284.5$ kJ/kmol **The heat of formation of CO is _____ kJ/kmol** – 109.5 + 109.5 – 678.5 + 678.5	**CO5**	**6**

	Ans: – 109.5		
4	Match the following: **Quantity** / **SI Unit** / **AE Unit** a) Temperature / 1) meter / i. Rankine b) Mass / 2) Kelvin / ii. pound c) Amount of substance / 3) kilogram / iii. foot d) Length / 4) gram-mole / iv. pound-mole a-2-i, b-3-ii, c-1-iv, d-4-iii a-2-i, b-3-iv, c-4-ii, d-1-iii a-2-i, b-3-ii, c-4-iv, d-1-iii a-4-i, b-3-ii, c-2-iv, d-1-iii Ans: a-2-i, b-3-ii, c-4-iv, d-1-iii	**CO1**	**1**
5	At 25 °C, an aqueous solution containing 35% H_2SO_4 has a specific gravity of 1.2563. A quantity of 35% H_2SO_4 solution that contains 195.5 kg of H_2SO_4 is needed. Calculate the required volume of the solution in liters. 445 l 420 l 470 l 438 l Ans: 445 l	**CO2**	**6**
6	A drug solution has a concentration of 50 mg/mL. What amount of drug is contained within 20.5 mL of the solution? 1.314 g 1.025 g 1.471 g 1.546 g Ans: 1.025 g	**CO2**	**6**
7	Gas leaving a fermenter at close to 1 atm pressure and 25°C has the following composition (mole%): 78.2% nitrogen, 19.2% oxygen, 2.6% carbon dioxide. Calculate the average molecular weight of the fermenter off-gas. 29.184 33.253 25.132 26.134 Ans: 29.184	**CO2**	**6**

8	Equal masses of CH_4 and H_2 are mixed in an empty container. The partial pressure of hydrogen in this container expressed as the fraction of total pressure is _____. 1/9 1/2 8/9 5/9 Ans: 8/9	**CO2**	**6**
9	Biomass is found to contain 50% C, 7% H, 33% O and 10% N by weight. If the molecular weight of the Biomass is 144, determine the molecular formula of the Biomass $C_8H_{10}NO_3$ $C_6H_5NO_3$ $C_6H_{10}NO_5$ $C_6H_{10}NO_3$ Ans: $C_6H_{10}NO_3$	**CO3**	**4**
10	A Biochemist is interested in preparing 1 liter 2N HCl solution. Find the grams of HCl needed to prepare the solution. Molecular weight of HCl is 36.5. 65 g 67 g 63 g 73 g Ans: 73 g	**CO2**	**4**
11	One hundred pounds of water is flowing through a pipe at the rate of 10 ft/s. what is the kinetic energy of this water in (ft)*(lb_f)? 135 (ft)*(lb_f) 141 (ft)*(lb_f) 155 (ft)*(lb_f) 122 (ft)*(lb_f) Ans: 155 (ft)*(lb_f)	**CO1**	**3**
12	The volume of a microbial culture is observed to increase according to the relation V (cm^3) = e^{3t} where t is in seconds. What is the unit of constant 3? cm 1/s cm^2 1/cm Ans: 1/s	**CO1**	**5**

13	A solution of $ZnBr_2$ in water contains 130 g salt per 100 mL solution at 293 K. The specific gravity of the solution is 2. Calculate the concentration of $ZnBr_2$ in weight percent. 40% 55% 65% 80% Ans: 65%	**CO2**	**6**
14	Which one is the most concentrated solution. 20 g of NaOH in 100 mL of water 150 g of NaOH in 200 mL of water 180 g of NaOH in 300 mL of water 250 g of NaOH in 600 mL of water Ans: 150 g of NaOH in 200 mL of water	**CO2**	**4**
15	Densities for _____ and _____do not change significantly at ordinary conditions with pressure. Gas, liquids Liquids, solids Solids, gas None of the mentioned Ans: Liquids, solids	**CO2**	**4**
16	Which one is not a dimension. Length Mass Time Kelvin Ans: Kelvin	**CO1**	**1**

17	_____ are the means of explicitly expressing the dimensions. Units Dimensions Significant figures Neither unit nor dimension Ans: Units	**CO1**	**1**
18	Real gases approach ideal behavior at High pressure and high temperature Low pressure and high temperature Low pressure and low temperature High pressure and low temperature Ans: Low pressure and high temperature	**CO2**	**1**
19	100 m^3 of a gas mixture containing 20% CO_2 is sent to an absorber in which 90% of CO_2 is absorbed. The temperature and pressure of the gas entering are the same as those of the gas leaving. What will be the pure – component volume of CO_2 in the gas leaving? 2 m^3 80 m^3 82 m^3 100 m^3 Ans: 2 m^3	**CO2**	**6**
20	What can be said about a set of data when its SD is small (but not zero)? The data are far apart. All of the data have the same value. The mean of the data can never be zero. The data are close together. Ans: The data are close together.	**CO2**	**1**
21	$Na_2SO_4.10H_2O$ crystals are formed by cooling 100 kg of 30% by weight aqueous solution of Na_2SO_4 (Mol.Wt 142). The final concentration of the solute in the solution is 10%. The weight of crystals is 22.33 32.2 45.35 58.65 Ans: 58.65	**CO3**	**6**

22	Which of the following statements is FALSE for the chemical equation given below in which nitrogen gas reacts with hydrogen gas to form ammonia gas assuming the reaction goes to completion? $N_2 + 3H_2 \longrightarrow 2NH_3$ The reaction of one mole of H_2 will produce 2/3 moles of NH_3. One mole of N_2 will produce two moles of NH_3. One molecule of nitrogen requires three molecules of hydrogen for complete reaction. The reaction of 14 g of nitrogen produces 27 g of ammonia. Ans: The reaction of 14 g of nitrogen produces 27 g of ammonia.	**CO4**	**5**
23	How many moles of $KBrO_3$ are required to prepare 0.0700 moles of Br_2 according to the reaction: $KBrO_3 + 5KBr + 6HNO_3$ ----> $6KNO_3 + 3Br_2 + 3H_2O$ 0.023 0.0732 0.0704 0.220 Ans: 0.023	**CO4**	**6**
24	The limiting reagent in a chemical reaction is one that: has the largest molar mass (formula weight). has the smallest molar mass (formula weight). has the smallest coefficient. is consumed completely. Ans: is consumed completely.	**CO4**	**1**
25	Yeast converts glucose to ethanol and carbon dioxide by glycolysis as per the following reaction: $C_6H_{12}O_6$ --------> $2C_2H_5OH + 2CO_2$. Assuming complete conversion, the amount of ethanol produced (in g) from 200 g of glucose is _____ 180 g 115 g 95 g 102g Ans: 102g	**CO4**	**6**
26	The degree of reduction for acetic acid ($C_2H_4O_2$) is _____ 2 8 4 3 Ans: 4	**CO4**	**3**

27	_____ are controlled or fixed during the experiment. Random errors Independent and Dependent Variables Dependent Variables Independent Variables Ans: Independent Variables	**CO2**	**1**
28	The deviation of an individual value from the mean is known as the _____. Systematic error Random error Standard deviation Residual Ans: Residual	**CO2**	**1**
29	Which of the following is not a unit of energy W*sec kg*m/sec N*m Joule Ans: kg*m/sec	**CO1**	**5**
30	Medical herbs are extracted with hexane in batch extractors. The herbal material contain 20% oil, 68% solids and 12% moisture. At the end of the extraction process, the cake is separated from the hexane – oil mixture. The cake analysis yields 0.8% oil, 88% solids, and 11.2% moisture. Find the percentage recovery of oil. 91% 84% 97% 94% Ans: 97%	**CO3**	**6**

Cognitive	Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation
Levels	1	2	3	4	5	6